

p2

Package in package 'Model'

p2

Version 1.0 Phase 1.0 Proposed

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Classes_MVC diagram

Class diagram in package 'p2'

Classes_MVC

Version 1.0

steamuser created on 8/10/2026. Last modified 8/10/2026

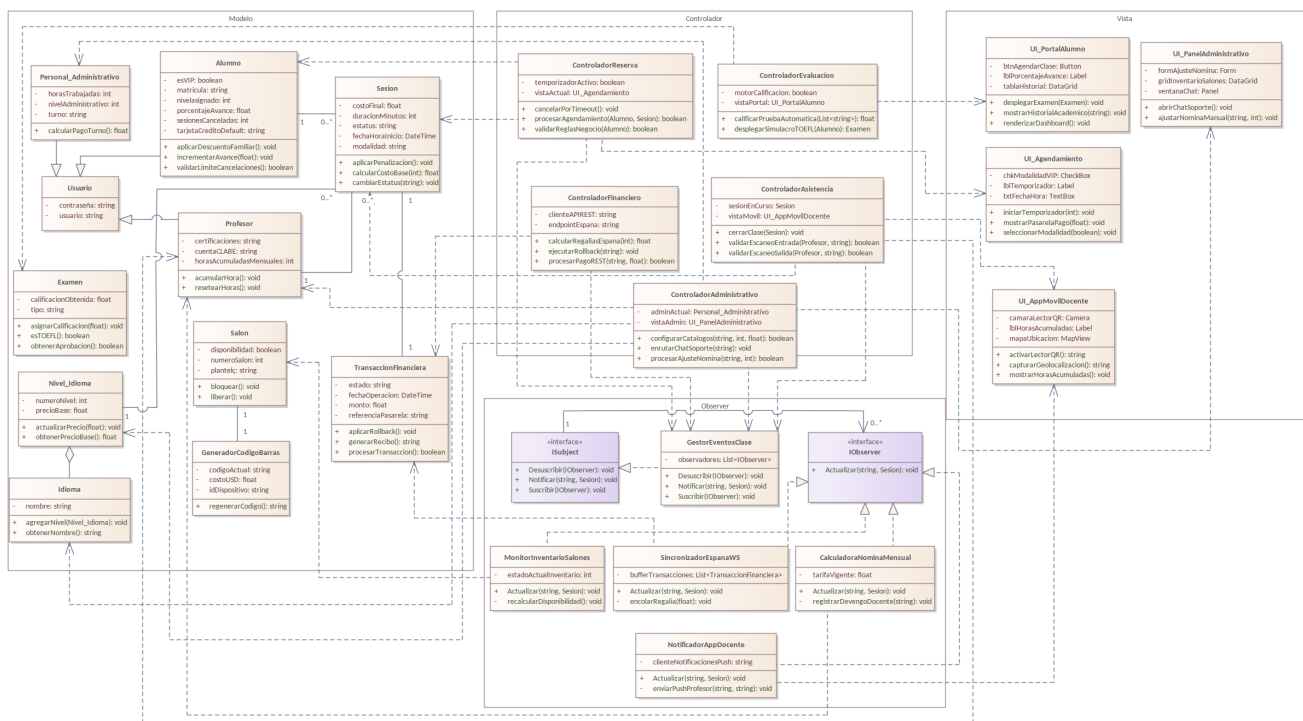


Figure 1: Clases_MVC

Controlador

Boundary in package 'p2'

Controlador

Version 1.0 Phase 1.0 Proposed

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Extends

Modelo

Boundary in package 'p2'

Modelo

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Extends

Observer

Boundary in package 'p2'

Observer
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Extends

Vista

Boundary in package 'p2'

Vista
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Extends

Alumno

Class in package 'p2'

Alumno
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OUTGOING STRUCTURAL RELATIONSHIPS

← Generalization from Alumno to Usuario

[Direction is 'Source -> Destination'.]

CONNECTORS

 **Dependency** Source -> Destination
From: ControladorReserva : Class, Public
To: Alumno : Class, Public

ATTRIBUTES

 esVIP : boolean Private

[Is static True. Containment is Not Specified.]

 matricula : string Private

ATTRIBUTES	
	[Is static True. Containment is Not Specified.]
◆ nivelasignado : int Private	[Is static True. Containment is Not Specified.]
◆ porcentajeAvance : float Private	[Is static True. Containment is Not Specified.]
◆ sesionesCanceladas : int Private	[Is static True. Containment is Not Specified.]
◆ tarjetaCreditoDefault : string Private	[Is static True. Containment is Not Specified.]

ASSOCIATIONS	
✎ Association (direction: Unspecified)	
Source: Public (Class) Alumno Cardinality: [1]	Target: Public (Class) Sesion Cardinality: [0..*]

OPERATIONS	
◆ aplicarDescuentoFamiliar () : void Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
◆ incrementarAvance (calificacion : float) : void Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
◆ validarLimiteCancelaciones () : boolean Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

CalculadoraNominaMensual

Class in package 'p2'

CalculadoraNominaMensual
Version 1.0 Phase 1.0 Proposed
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OUTGOING STRUCTURAL RELATIONSHIPS	
← Realization from CalculadoraNominaMensual to IObserver	[Direction is 'Source -> Destination'.]

CONNECTORS

 **Dependency** Source -> Destination
 From: CalculadoraNominaMensual : Class, Public
 To: Profesor : Class, Public

ATTRIBUTES

 tarifaVigente : float Private

[Is static True. Containment is Not Specified.]

OPERATIONS

 Actualizar (tipoEvento : string , s : Sesión) : void Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 registrarDevengoDocente (idProfesor : string) : void Private
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ControladorAdministrativo

Class in package 'p2'

ControladorAdministrativo
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CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: UI_PanelAdministrativo : Class, Public

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: Profesor : Class, Public

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: Idioma : Class, Public

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: Personal_Administrativo : Class, Public

CONNECTORS

**Dependency** Source -> Destination

From: ControladorAdministrativo : Class, Public

To: GestorEventosClase : Class, Public

**Dependency** Source -> Destination

From: ControladorAdministrativo : Class, Public

To: Nivel_Idioma : Class, Public

ATTRIBUTES



adminActual : Personal_Administrativo Private

[Is static True. Containment is Not Specified.]



vistaAdmin : UI_PanelAdministrativo Private

[Is static True. Containment is Not Specified.]

OPERATIONS



configurarCatalogos (idioma : string , nivel : int , precio : float) : boolean Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]



enrutarChatSoporte (idUserario : string) : void Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]



procesarAjusteNomina (idProfesor : string , horasExtra : int) : boolean Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ControladorAsistencia

Class in package 'p2'

ControladorAsistencia
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CONNECTORS

**Dependency** Source -> Destination

From: ControladorAsistencia : Class, Public

To: UI_AppMovilDocente : Class, Public

**Dependency** Source -> Destination

From: ControladorAsistencia : Class, Public

To: GestorEventosClase : Class, Public

CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAsistencia : Class, Public
 To: Profesor : Class, Public

 **Dependency** Source -> Destination
 From: ControladorAsistencia : Class, Public
 To: Sesion : Class, Public

ATTRIBUTES

 sesionEnCurso : Sesion Private [Is static True. Containment is Not Specified.]

 vistaMovil : UI_AppMovilDocente Private [Is static True. Containment is Not Specified.]

OPERATIONS

 cerrarClase (s : Sesion) : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 validarEscaneoEntrada (p : Profesor , qrLeido : string) : boolean Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 validarEscaneoSalida (p : Profesor , qrLeido : string) : boolean Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ControladorEvaluacion

Class in package 'p2'

ControladorEvaluacion
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CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorEvaluacion : Class, Public
 To: UI_PortalAlumno : Class, Public

 **Dependency** Source -> Destination
 From: ControladorEvaluacion : Class, Public
 To: Examen : Class, Public

ATTRIBUTES	
 motorCalificacion : boolean Private	[Is static True. Containment is Not Specified.]
 vistaPortal : UI_PortalAlumno Private	[Is static True. Containment is Not Specified.]

OPERATIONS	
 calificarPruebaAutomatica (respuestas : List<string>) : float Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 desplegarSimulacroTOEFL (a : Alumno) : Examen Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ControladorFinanciero

Class in package 'p2'

ControladorFinanciero
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CONNECTORS	
 Dependency Source -> Destination From: ControladorFinanciero : Class, Public To: GestorEventosClase : Class, Public	
 Dependency Source -> Destination From: ControladorFinanciero : Class, Public To: TransaccionFinanciera : Class, Public	

ATTRIBUTES	
 clienteAPIREST : string Private	[Is static True. Containment is Not Specified.]
 endpointEspana : string Private	[Is static True. Containment is Not Specified.]

OPERATIONS	
 calcularRegaliasEspana (mes : int) : float Public	

OPERATIONS
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 ejecutarRollback (referenciaTransaccion : string) : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 procesarPagoREST (tarjeta : string , monto : float) : boolean Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ControladorReserva

Class in package 'p2'

ControladorReserva
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CONNECTORS
 Dependency Source -> Destination From: ControladorReserva : Class, Public To: GestorEventosClase : Class, Public
 Dependency Source -> Destination From: ControladorReserva : Class, Public To: UI_Agendamiento : Class, Public
 Dependency Source -> Destination From: ControladorReserva : Class, Public To: Sesion : Class, Public
 Dependency Source -> Destination From: ControladorReserva : Class, Public To: Alumno : Class, Public

ATTRIBUTES
 temporizadorActivo : boolean Private [Is static True. Containment is Not Specified.]
 vistaActual : UI_Agendamiento Private [Is static True. Containment is Not Specified.]

OPERATIONS
 cancelarPorTimeout () : void Public

OPERATIONS	
	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 procesarAgendamiento (a : Alumno , s : Sesion) : boolean Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 validarReglasNegocio (a : Alumno) : boolean Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Examen

Class in package 'p2'

Examen

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CONNECTORS	
 Dependency Source -> Destination From: ControladorEvaluacion : Class, Public To: Examen : Class, Public	

ATTRIBUTES	
 calificacionObtenida : float Private	[Is static True. Containment is Not Specified.]
 tipo : string Private	[Is static True. Containment is Not Specified.]

OPERATIONS	
 asignarCalificacion (calificacion : float) : void Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 esTOEFL () : boolean Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 obtenerAprobacion () : boolean Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Class in package 'p2'

GeneradorCodigoBarras
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ATTRIBUTES	
 codigoActual : string Private	[Is static True. Containment is Not Specified.]
 costoUSD : float Private	[Is static True. Containment is Not Specified.]
 idDispositivo : string Private	[Is static True. Containment is Not Specified.]
ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public (Class) Salon Cardinality: [1]	Target: Public (Class) GeneradorCodigoBarras Cardinality: [1]
OPERATIONS	
 regenerarCodigo () : string Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

GestorEventosClase

Class in package 'p2'

GestorEventosClase
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OUTGOING STRUCTURAL RELATIONSHIPS	
 Realization from GestorEventosClase to ISubject	[Direction is 'Source -> Destination'.]
CONNECTORS	
 Dependency Source -> Destination From: ControladorReserva : Class, Public To: GestorEventosClase : Class, Public	

CONNECTORS

 **Dependency** Source -> Destination
From: ControladorFinanciero : Class, Public
To: GestorEventosClase : Class, Public

 **Dependency** Source -> Destination
From: ControladorAsistencia : Class, Public
To: GestorEventosClase : Class, Public

 **Dependency** Source -> Destination
From: ControladorAdministrativo : Class, Public
To: GestorEventosClase : Class, Public

ATTRIBUTES

 observadores : List<IObserver> Private

[Is static True. Containment is Not Specified.]

OPERATIONS

 Desuscribir (o : IObserver) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 Notificar (tipoEvento : string , s : Sesion) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 Suscribirse (o : IObserver) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Idioma

Class in package 'p2'

Idioma

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OUTGOING STRUCTURAL RELATIONSHIPS

 Aggregation from Idioma to Nivel_Idioma

[Direction is 'Source -> Destination'.]

CONNECTORS

CONNECTORS

 **Dependency** Source -> Destination
From: ControladorAdministrativo : Class, Public
To: Idioma : Class, Public

ATTRIBUTES

 nombre : string Private
[Is static True. Containment is Not Specified.]

OPERATIONS

 agregarNivel (nivel : Nivel_Idioma) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 obtenerNombre () : string Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

MonitorInventarioSalones

Class in package 'p2'

MonitorInventarioSalones
Version 1.0 Phase 1.0 Proposed
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OUTGOING STRUCTURAL RELATIONSHIPS

 Realization from MonitorInventarioSalones to IObserver
[Direction is 'Source -> Destination'.]

CONNECTORS

 **Dependency** Source -> Destination
From: MonitorInventarioSalones : Class, Public
To: Salon : Class, Public

ATTRIBUTES

 estadoActualInventario : int Private
[Is static True. Containment is Not Specified.]

OPERATIONS

 Actualizar (tipoEvento : string , s : Sesión) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

OPERATIONS

💎 recalcularDisponibilidad () : void Private
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Nivel_Idioma

Class in package 'p2'

Nivel_Idioma

Version 1.0 Phase 1.0 Proposed

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INCOMING STRUCTURAL RELATIONSHIPS

➡ Aggregation from Idioma to Nivel_Idioma
[Direction is 'Source -> Destination'.]

CONNECTORS

🔗 **Dependency** Source -> Destination
From: ControladorAdministrativo : Class, Public
To: Nivel_Idioma : Class, Public

ATTRIBUTES

💎 numeroNivel : int Private
[Is static True. Containment is Not Specified.]

💎 precioBase : float Private
[Is static True. Containment is Not Specified.]

ASSOCIATIONS

🔗 Association (direction: Unspecified)
Source: Public (Class) Nivel_Idioma
Cardinality: [1]
Target: Public (Class) Sesion
Cardinality: [0..*]

OPERATIONS

💎 actualizarPrecio (nuevoPrecio : float) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

💎 obtenerPrecioBase () : float Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

OPERATIONS

NotificadorAppDocente

Class in package 'p2'

NotificadorAppDocente
Version 1.0 Phase 1.0 Proposed
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OUTGOING STRUCTURAL RELATIONSHIPS

← Realization from NotificadorAppDocente to IObservable

[Direction is 'Source -> Destination'.]

CONNECTORS

↗ **Dependency** Source -> Destination

From: NotificadorAppDocente : Class, Public

To: UI_AppMovilDocente : Class, Public

ATTRIBUTES

◆ clienteNotificacionesPush : string Private

[Is static True. Containment is Not Specified.]

OPERATIONS

◆ Actualizar (tipoEvento : string , s : Sesion) : void Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

◆ enviarPushProfesor (idProfesor : string , mensaje : string) : void Private

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Personal_Administrativo

Class in package 'p2'

Personal_Administrativo
Version 1.0 Phase 1.0 Proposed
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OUTGOING STRUCTURAL RELATIONSHIPS

OUTGOING STRUCTURAL RELATIONSHIPS

← Generalization from Personal_Administrativo to Usuario

[Direction is 'Source -> Destination'.]

CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: Personal_Administrativo : Class, Public

ATTRIBUTES

 horasTrabajadas : int Private

[Is static True. Containment is Not Specified.]

 nivelAdministrativo : int Private

[Is static True. Containment is Not Specified.]

 turno : string Private

[Is static True. Containment is Not Specified.]

OPERATIONS

 calcularPagoTurno () : float Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Profesor

Class in package 'p2'

Profesor
 Version 1.0 Phase 1.0 Proposed
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OUTGOING STRUCTURAL RELATIONSHIPS

← Generalization from Profesor to Usuario

[Direction is 'Source -> Destination'.]

CONNECTORS

 **Dependency** Source -> Destination
 From: CalculadoraNominaMensual : Class, Public
 To: Profesor : Class, Public

CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: Profesor : Class, Public

 **Dependency** Source -> Destination
 From: ControladorAsistencia : Class, Public
 To: Profesor : Class, Public

ATTRIBUTES

 certificaciones : string Private [Is static True. Containment is Not Specified.]

 cuentaCLABE : string Private [Is static True. Containment is Not Specified.]

 horasAcumuladasMensuales : int Private [Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) Profesor
 Cardinality: [1]

Target: Public (Class) Sesion
 Cardinality: [0..*]

OPERATIONS

 acumularHora () : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 resetearHoras () : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Salon

Class in package 'p2'

Salon
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CONNECTORS

CONNECTORS

 **Dependency** Source -> Destination
 From: MonitorInventarioSalones : Class, Public
 To: Salon : Class, Public

ATTRIBUTES

 disponibilidad : boolean Private [Is static True. Containment is Not Specified.]

 numeroSalon : int Private [Is static True. Containment is Not Specified.]

 plantelç : string Private [Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) Salon
 Cardinality: [1]

Target: Public (Class) GeneradorCodigoBarras
 Cardinality: [1]

OPERATIONS

 bloquear () : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 liberar () : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Sesion

Class in package 'p2'

Sesion
 Version 1.0 Phase 1.0 Proposed
 steamuser created on 8/10/2026. Last modified 8/10/2026

CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorReserva : Class, Public
 To: Sesion : Class, Public

OPERATIONS

- ◆ aplicarPenalizacion () : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
- ◆ calcularCostoBase (nivel : int) : float Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
- ◆ cambiarEstatus (nuevoEstatus : string) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

SincronizadorEspanaWS

Class in package 'p2'

SincronizadorEspanaWS
Version 1.0 Phase 1.0 Proposed
steamuser created on 8/10/2026. Last modified 8/10/2026

OUTGOING STRUCTURAL RELATIONSHIPS

- ← Realization from SincronizadorEspanaWS to IObserver
[Direction is 'Source -> Destination'.]

CONNECTORS

- ↗ **Dependency** Source -> Destination
From: SincronizadorEspanaWS : Class, Public
To: TransaccionFinanciera : Class, Public

ATTRIBUTES

- ◆ bufferTransacciones : List<TransaccionFinanciera> Private
[Is static True. Containment is Not Specified.]

OPERATIONS

- ◆ Actualizar (tipoEvento : string , s : Sesion) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
- ◆ encolarRegalia (monto : float) : void Private
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

TransaccionFinanciera

Class in package 'p2'

TransaccionFinanciera
Version 1.0 Phase 1.0 Proposed
steamuser created on 8/10/2026. Last modified 8/10/2026

CONNECTORS

 **Dependency** Source -> Destination
From: SincronizadorEspanaWS : Class, Public
To: TransaccionFinanciera : Class, Public

 **Dependency** Source -> Destination
From: ControladorFinanciero : Class, Public
To: TransaccionFinanciera : Class, Public

ATTRIBUTES

 estado : string Private [Is static True. Containment is Not Specified.]

 fechaOperacion : DateTime Private [Is static True. Containment is Not Specified.]

 monto : float Private [Is static True. Containment is Not Specified.]

 referenciaPasarela : string Private [Is static True. Containment is Not Specified.]

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public (Class) Sesion
Cardinality: [1]

Target: Public (Class) TransaccionFinanciera
Cardinality: [1]

OPERATIONS

 aplicarRollback () : void Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 generarRecibo () : string Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 procesarTransaccion () : boolean Public [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

UI_Agendamiento

Class in package 'p2'

UI_Agendamiento

Version 1.0 Phase 1.0 Proposed

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CONNECTORS

 **Dependency** Source -> Destination
From: ControladorReserva : Class, Public
To: UI_Agendamiento : Class, Public

ATTRIBUTES

 chkModalidadVIP : CheckBox Private
[Is static True. Containment is Not Specified.]

 lblTemporizador : Label Private
[Is static True. Containment is Not Specified.]

 txtFechaHora : TextBox Private
[Is static True. Containment is Not Specified.]

OPERATIONS

 iniciarTemporizador (minutos : int) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 mostrarPasarelaPago (montoFinal : float) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 seleccionarModalidad (esVIP : boolean) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

UI_AppMovilDocente

Class in package 'p2'

UI_AppMovilDocente

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CONNECTORS

CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAsistencia : Class, Public
 To: UI_AppMovilDocente : Class, Public

 **Dependency** Source -> Destination
 From: NotificadorAppDocente : Class, Public
 To: UI_AppMovilDocente : Class, Public

ATTRIBUTES

 camaraLectorQR : Camera Private
 [Is static True. Containment is Not Specified.]

 lblHorasAcumuladas : Label Private
 [Is static True. Containment is Not Specified.]

 mapaUbicacion : MapView Private
 [Is static True. Containment is Not Specified.]

OPERATIONS

 activarLectorQR () : string Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 capturarGeolocalizacion () : string Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

 mostrarHorasAcumuladas () : void Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

UI_PanelAdministrativo

Class in package 'p2'

UI_PanelAdministrativo
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CONNECTORS

 **Dependency** Source -> Destination
 From: ControladorAdministrativo : Class, Public
 To: UI_PanelAdministrativo : Class, Public

ATTRIBUTES	
 formAjusteNomina : Form Private	[Is static True. Containment is Not Specified.]
 gridInventarioSalones : DataGrid Private	[Is static True. Containment is Not Specified.]
 ventanaChat : Panel Private	[Is static True. Containment is Not Specified.]

OPERATIONS	
 abrirChatSoporte () : void Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
 ajustarNominaManual (idProfesor : string , horasExtra : int) : void Public	[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

UI_PortalAlumno

Class in package 'p2'

UI_PortalAlumno
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CONNECTORS	
 Dependency Source -> Destination	
From: ControladorEvaluacion : Class, Public	
To: UI_PortalAlumno : Class, Public	

ATTRIBUTES	
 btnAgendarClase : Button Private	[Is static True. Containment is Not Specified.]
 lblPorcentajeAvance : Label Private	[Is static True. Containment is Not Specified.]
 tablaHistorial : DataGrid Private	[Is static True. Containment is Not Specified.]

OPERATIONS	
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OPERATIONS

-  desplegarExamen (ex : Examen) : void Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
-  mostrarHistorialAcademico (matricula : string) : void Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]
-  renderizarDashboard () : void Public
 [Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

Usuario

Class in package 'p2'

Usuario
 Version 1.0 Phase 1.0 Proposed
 steamuser created on 8/10/2026. Last modified 8/10/2026

INCOMING STRUCTURAL RELATIONSHIPS

-  Generalization from Profesor to Usuario
 [Direction is 'Source -> Destination'.]
-  Generalization from Alumno to Usuario
 [Direction is 'Source -> Destination'.]
-  Generalization from Personal_Administrativo to Usuario
 [Direction is 'Source -> Destination'.]

ATTRIBUTES

-  contraseña : string Private
 [Is static True. Containment is Not Specified.]
-  usuario : string Private
 [Is static True. Containment is Not Specified.]

IObserver

Interface in package 'p2'

IObserver
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INCOMING STRUCTURAL RELATIONSHIPS

⇒ Realization from SincronizadorEspanaWS to IObserver

[Direction is 'Source -> Destination'.]

⇒ Realization from CalculadoraNominaMensual to IObserver

[Direction is 'Source -> Destination'.]

⇒ Realization from NotificadorAppDocente to IObserver

[Direction is 'Source -> Destination'.]

⇒ Realization from MonitorInventarioSalones to IObserver

[Direction is 'Source -> Destination'.]

ASSOCIATIONS

✎ Association (direction: Source -> Destination)

Source: Public (Interface) ISubject
Cardinality: [1]Target: Public (Interface) IObserver
Cardinality: [0..*]

OPERATIONS

💎 Actualizar (tipoEvento : string , s : Sesion) : void Public

[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

ISubject

Interface in package 'p2'

ISubject

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INCOMING STRUCTURAL RELATIONSHIPS

⇒ Realization from GestorEventosClase to ISubject

[Direction is 'Source -> Destination'.]

ASSOCIATIONS

✎ Association (direction: Source -> Destination)

Source: Public (Interface) ISubject
Cardinality: [1]Target: Public (Interface) IObserver
Cardinality: [0..*]

OPERATIONS

◆ Desuscribir (o : IObserver) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

◆ Notificar (tipoEvento : string , s : Sesion) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]

◆ Suscribirse (o : IObserver) : void Public
[Is static False. Is abstract False. Is return array False. Is query False. Is synchronized False.]